Q) Create structure variable student and estimate size of structure manually. Then write a C programs to find size of structure using sizeof programs. Now observe both the outputs. Were they same or different? Justify your answer.

Structure definition:

```
Struct student
{
    int rollno;
    char grade;
    float marks;
};
```

Size estimation

int roll = 4 bytes
char grade = 1 byte
float marks = 4 bytes.
Size = 4+1+4 = 9 bytes

C program using size of

```
#include <stdio.h>
struct student
{
    int rollno;
    char grade;
    float marks;
};
int main()
{
  struct student s;
  printf("size = %d 20 bytes \n sizeof(s));
  return 0;
}
```

o/p size = 12 bytes.

∴ the O/P are different because Compiler adds Padding btw structure members so that data is stored at memory addresses that are efficient for Processor to access. Although actual data occupies 6 bytes, Compiler inserts 3 extra bytes, making it 12 bytes.